Al-Balqa Applied University

جامعة البلقاء التطبيقية

PART I

Choose the most correct answer for the following questions and transfer it in CAPITAL letters to table on the cover page.

Q1-What should you do if you accidentally spill a chemical on yourself?

A. Wait until the end of the experiment to clean it up
B. Clean it up immediately using the appropriate procedure and notify your supervisor.
C. Ignore it, as small spills are not harmful.
D. Ask someone else to clean it up for you

Q2-Which of the following is true about eating or drinking in the lab?

A. It is allowed as long as you are careful not to spill
B. It is allowed during non-experimental periods
C. It is allowed in designated areas of the lab
D. It is strictly forbidden at all times

Q3-When using a Bunsen burner, which statement is correct?

A. It's okay to leave the Bunsen burner unattended while it's lit
B. Always point the open end of the tube away from yourself and others.
C. Place the Bunsen burner near flammable materials for easy access.
D. It's not necessary to have a fire extinguisher nearby while using the burner.

Q4-What is the mass of a cubic block (cube) of silicon of side 7.663 cm length, given that density of silicon(Si) is 2.336 g/cm³?

A. 1051.2g B. 17.9 g C. 192.6g D. 2102.4g

Q5-Which one of the following salts is water soluble?

A. $BaSO_4$ B. $CaCO_3$ C. K_3PO_4 D. $Fe(OH)_2$

Q6-What is the function of a capillary tube in a boiling point determination?

A. To measure the volume of the liquid.
B. To provide a nucleation site where you can easily observe the bubble formation.
C. To condense the vapor.
D. To increase the vapor pressure of liquid.

Q7-A student measures the boiling point of liquid water to be 102°C at an atmospheric pressure of 1.0 atm. What can be concluded?

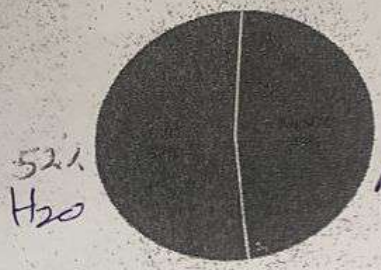
A. The liquid water contains soluble impurities.
B. The liquid water contains insoluble impurities.
C. The liquid water is pure.
D. A&B

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Q8-Calculate water of crystallization (X) for the salt which shown beside knowing that M.m of ($MgSO_4$ =120.3g/mol and H_2O= 18g/mol)

A. 7
B. 8
C. 9
D. 5



Answer the following questions and *transfer your answer to the table on the cover page*

Q		Answer
1	What is the name of the glassware in the following picture? 	volimtric flask
2	Two liquids, A and B, have densities 0.75 grams per milliliter and 1.14 grams per milliliter, respectively. When both liquids are poured into a container, one liquid floats on top of the other. Which liquid is on top?	A
3	What physical observation do you observe when C_6H_{12} is mixed with water?	purple color
4	If a hydrated salt has the formula $CuSO_4 \cdot 5H_2O$, how many moles of water are present per 1.5 mole of $CuSO_4$?	
5	If a hydrated salt is not completely dehydrated, what potential error might occur on the value of X?	
6	Which piece of equipment is used to place the hydrated salt in, ready to be heated?	
7	A hydrate of magnesium chloride is heated, and the following data is collected: mass of crucible = 22.130 grams mass of crucible + hydrate = 25.290 grams mass of crucible and contents after heating = 23.491 grams What is the % of the anhydrous $MgCl_2$ in the sample ?	56.9%

Good LUCK